

- 4 Fig. 4.1 is a photomicrograph of a cross-section of a tubular structure in the kidney made from epithelial cells.



Fig. 4.1

- (a) The actual length of epithelial cell **A** along the line **P–Q** is $35\ \mu\text{m}$.

Calculate the magnification of the image shown in Fig. 4.1. Write down the formula and use it to make your calculation. Show your working.

formula

magnification $\square$ [2]

- (b)** Some epithelial cells in the kidney release the protein vascular endothelial growth factor (VEGF). This protein is a cell signalling molecule that stimulates cell division in endothelial cells in blood vessels.

(i) State what occurs during interphase to prepare a cell for division.

.....

.....

.....

.....

..... [2]

(ii) Explain how a cell signalling molecule, such as VEGF, can lead to a response in a cell.

.....

.....

.....

..... [2]

- (c)** Uncontrolled cell division may result in a tumour. Tumour cells in the kidney respond to VEGF.

Kidney cancer can be treated with monoclonal antibodies. These monoclonal antibodies bind to VEGF.

Outline the hybridoma method for the production of monoclonal antibodies that will target the VEGF protein.

.....

.....

.....

.....

.....

..... [4]

- (d)** Monoclonal antibodies used as a treatment need to be given more than once. Repeated treatment can cause side effects to the person or can become less effective.

Suggest why repeated treatment with monoclonal antibodies may have these effects.

.....

.....

.....

..... [1]